

whereas payments for the same may be received six months later and hence may not be recorded in BOP of that year. Therefore, in order to balance the BOP account an entry for *errors and omissions* is included.

Even though the BOP always balances, we can talk about an overall BOP surplus or deficit. Overall BOP balance is obtained by adding the current and capital account balances. A country has a BOP surplus in the following cases: (i) there is current account surplus as well as capital account surplus; (ii) there is a current account deficit but a capital account surplus that is larger in magnitude; (iii) there is a current account surplus and a relatively smaller capital account deficit. There would be a BOP deficit, either when there is a deficit on both current and capital accounts or when there is a relatively large deficit in either one of the current or capital accounts.

In the case of a BOP surplus there is an increase in the foreign exchange reserves of the country. In the case of a deficit, the stock of foreign exchange reserves is depleted. Let us discuss this issue in details.

Change in Foreign Exchange Reserves

Apart from transactions made by private and government agents, the capital account of the BOP also includes *official reserve transactions* that involve change in the stock of foreign exchange reserves held by the central bank. When there is a BOP surplus, the country experiences a net inflow of foreign exchange, leading to a corresponding increase in foreign exchange reserves. In the case of a BOP deficit, there is net outflow of foreign exchange and a corresponding reduction in foreign exchange reserves. Thus,

$$\text{BOP Balance} = \text{Current Account} + \text{Capital Account (including errors and omissions)} + \text{Official Reserve Transactions} = 0$$

In case a country has a current account deficit (of say, – \$15,000), but a capital account surplus that is smaller in magnitude (say, \$12,000), then it has a BOP deficit (of –\$3000) that leads to a decrease in foreign exchange reserves (by \$3000).

Note that a BOP surplus is recorded with a positive sign in the BOP, while the corresponding *increase* in reserves has a *negative* sign, so that the two add up to zero. Therefore in BOP accounts an increase in foreign exchange reserves appears with a negative sign whereas a decrease in reserves has a positive sign.

4.3.2 Exchange Rate

The exchange rate is an important concept in an open economy such as India, that has transactions involving foreign goods, services and assets, whose prices are denominated in terms of foreign currencies. Throughout the following discussion we assume that the domestic country is India and the foreign country is the USA.

Demand for foreign exchange arises whenever domestic agents purchase foreign goods, services or assets (i.e., imports and capital outflows); whereas, sale of domestic goods, services and assets to foreigners (i.e., exports and capital

inflows) constitutes the primary source of supply of foreign exchange. The market value of the exchange rate is determined by demand and supply in the foreign exchange market. However, often the official exchange rate may differ from its market value as you will see from our discussion on alternate exchange rate regimes. First we shall learn about various exchange rate concepts.

Nominal Exchange Rates

The nominal exchange rate 'e' can be defined as the number of units of domestic currency (Rupee) required for purchasing one unit of foreign currency (dollar). For example, if Rs. 80 is required to buy one dollar, then the exchange rate $e = \text{Rs. 80 per dollar}$. Similarly, the exchange rate of the Indian rupee can also be defined in terms of other currencies, e.g., yen, euro, pound, etc.

When the exchange rate 'e' rises (say, from Rs. 80 to Rs.82 per dollar), there is an increase in the price of dollars in terms of rupees and there is **nominal depreciation** of the rupee vis-à-vis dollar (and **nominal appreciation** of the dollar vis-à-vis rupee). When 'e' falls (e.g., from Rs. 80 to Rs. 78 per dollar), the dollar becomes cheaper in terms of rupees and there is a **nominal appreciation** of the rupee (and **nominal depreciation** of the dollar vis-à-vis rupee).

Real Exchange Rate

While nominal exchange rates measure the rate of exchange between domestic and foreign currencies, the real exchange rate captures the rate at which domestic goods are exchanged for foreign goods. It is the price of foreign goods in terms of domestic goods and is computed using the nominal exchange rate and prices of a comparable basket of domestic and foreign goods as below:

$$r = (e p_f) / p_d \quad (4.6)$$

where 'e' is the nominal exchange rate (rupees per dollar), p_d denotes the domestic price level (in rupees) and p_f denotes the foreign price level (in dollars). ' $e p_f$ ' is the price in rupees of a foreign basket of goods and ' p_d ' is the price (in Rupees) of a comparable domestic basket.

The real exchange rate between rupee and dollar reflects the price of goods produced in the USA relative to those produced in India. A rise in r means American goods become relatively more expensive vis a vis Indian goods and there is **real depreciation** of the rupee; whereas a reduction in r makes Indian goods relatively more expensive and there is **real appreciation** of the rupee. The real exchange rate is often used as an index of a country's international price competitiveness. In case a country faces real appreciation of its currency, domestic goods become more expensive and this can lead to a reduction in demand for exports. In contrast, a real depreciation of the currency can boost export demand.

When domestic and foreign prices remain unchanged (that is, $\frac{p_f}{p_d}$ is constant), a nominal depreciation ('e' rises) of the domestic currency causes a corresponding real depreciation ('r' rises). In this case the nominal and real exchange rates

move in the same direction. However, prices may change simultaneously with exchange rates; e.g., an appreciation in nominal exchange rates ('e' falls), along with a large fall in domestic prices (p_d falls) may lead to a real depreciation (r rises). So, 'e' and 'r' may also move in opposite directions.

Exchange Rate Regimes

The exchange rate regime of a country determines how its exchange rate is determined.

Under a ***flexible (or floating) exchange rate regime***, the value of 'e' is determined by the demand and supply of foreign exchange, without any government intervention. In this case the currency is on a clean float or pure float. That means the exchange rate is fully flexible and adjusts continuously to equate market demand and supply of foreign exchange, so there is no role for intervention by the central bank and hence no need of foreign exchange reserves. In this case BOP balance is attained via exchange rate fluctuations. That is, in case there is a BOP surplus (supply of foreign currency exceeds demand), the exchange rate appreciates (e falls, so that given prices, r falls, so foreign goods are relatively cheaper), X falls, M rises and NX falls (the current account deficit widens and BOP surplus is reduced), till BOP equilibrium is restored. Floating exchange rates lead to considerable volatility in exchange rates that may be unfavorable for exporters and importers and may have adverse impacts on domestic production, investment and employment. As such, 'pure floats' are rarely observed in practice.

In most countries central banks maintain a stock of foreign exchange reserves and intervene and try to maintain the value of exchange rate within a certain band. Such exchange rate arrangements, widely prevalent across countries, are called a 'managed float' or dirty float. ***Intervention*** refers to the central bank buying or selling foreign exchange in an attempt to influence the exchange rate. In a managed float, the currency is allowed to float within a certain 'target zone' and the central bank stands ready to intervene (e.g., buy or sell dollars against rupees) whenever the exchange rate appears to breach the limits of this band.

At the other extreme we have the system of fixed exchange rate, wherein the exchange rate is not freely floating, but is fixed by the government. An important function of central banks in a fixed exchange rate system is to intervene in foreign exchange markets to keep the price of foreign exchange fixed at the pre-announced level. For this, central banks have to maintain adequate stock of foreign exchange reserves.

Under a fixed exchange rate system, the fixed parity announced by the central bank may be revised at times; in this case a decrease in 'e' due to official intervention is called ***revaluation*** while an increase in 'e' by official intervention is called ***devaluation***. In a world with few restrictions on international transactions in goods and assets, maintaining a fixed exchange rate poses an